



Motilal Nehru National Institute of Technology Allahabad, Prayagraj

Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning Lab

(CSN12601)

Lab Assignment – 4

S. N.	Questions
1.	Create a NumPy array and demonstrate common array manipulation operations, including append() , insert() , pop() , remove() , index() , sort() , and reverse() . The program should clearly show the effect of each operation on the array through appropriate print statements and comments, ensuring clarity of functionality and output.
2.	Write a Python program that demonstrates array copying using three different approaches : a) Assignment operator (reference copying) b) Shallow copy method c) Deep copy method The program should clearly illustrate the differences between these approaches by modifying the copied arrays and showing how the changes affect (or do not affect) the original array.
3.	Write a Python program that demonstrates key functionalities of the math and cmath modules. Your program should include clear examples of functions such as: • math.acos() , math.ceil() , math.exp() , math.gcd() • cmath.tan(x) , cmath.log10(x) , cmath.isclose() The program should take appropriate inputs (where required), display well-formatted outputs, and briefly indicate what each function does through comments or print statements.
4.	Write a Python program to create a calculator using functions , where each arithmetic operation (addition, subtraction, multiplication, division, etc.) is implemented as a separate function. The program should accept user input, perform the selected operation, handle invalid inputs appropriately, and display the result clearly.